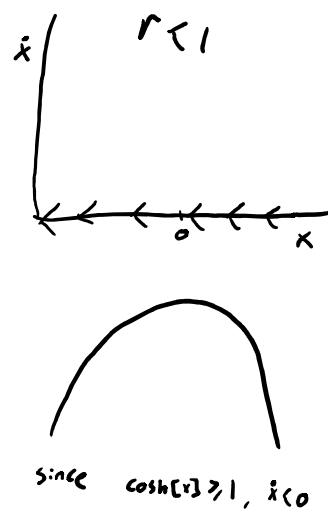
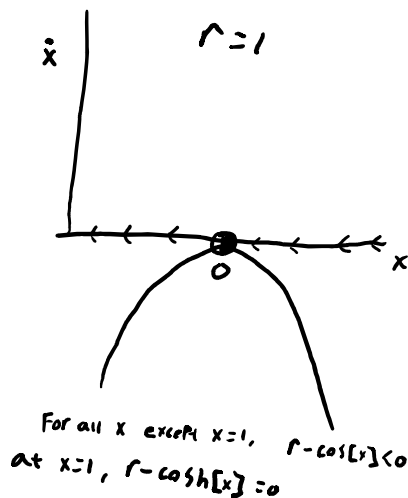
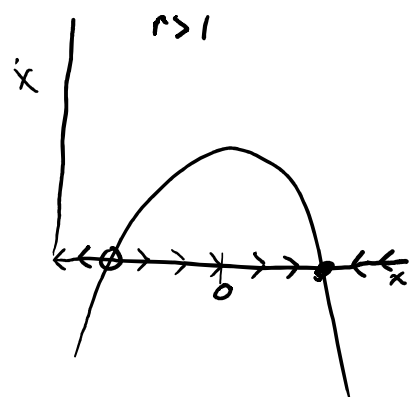


3.1.2

$$\dot{x} = r - \cosh[x]$$

There are three qualitatively different scenarios

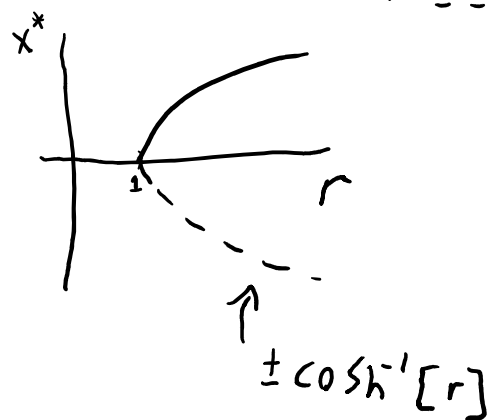


To show that a saddle-node bifurcation occurs, let's find how the fixed point solution, x^* , varies with r .

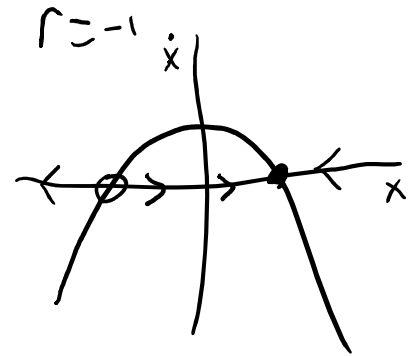
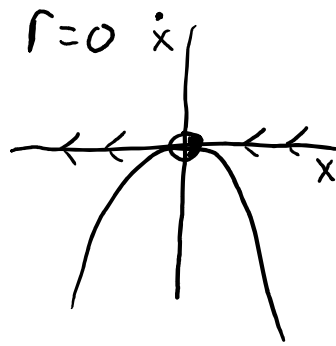
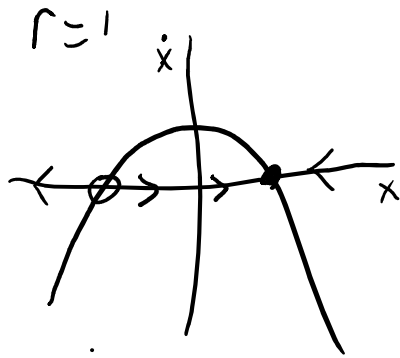
$$\dot{x} = 0 = r - \cosh[x^*] \Rightarrow r = \cosh[x^*]$$

So, For $r \geq 1$, there is a solution at $x^* = \pm \cosh^{-1}[r]$

This plot reveals a saddle-node bifurcation at $r=1$



3.1.5 a) $\dot{x} = r^2 - x^2$

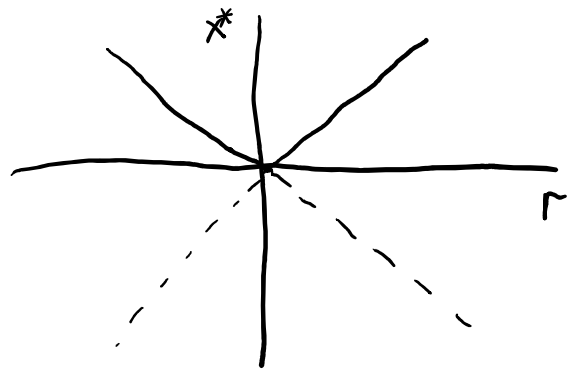


$$\dot{x} = 0 = r^2 - x^{*2}$$

$$\Rightarrow r^2 = x^{*2}$$

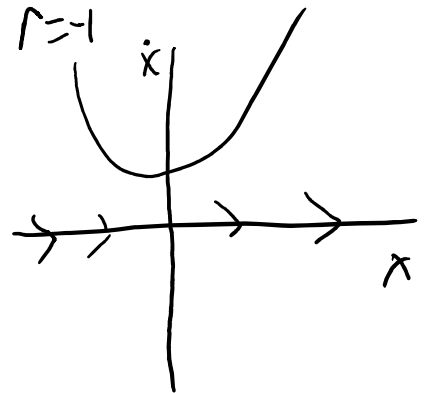
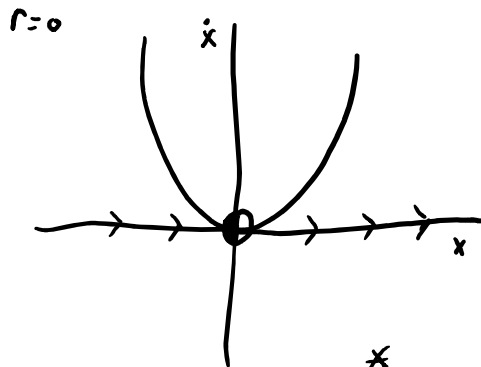
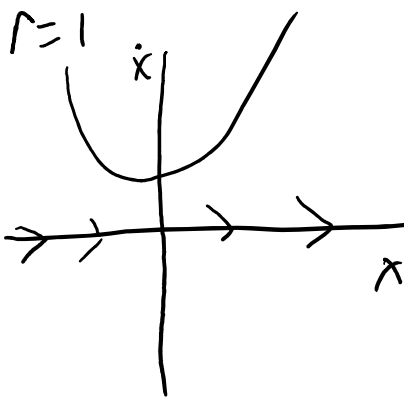
So,

$$x^* = \pm |r|$$



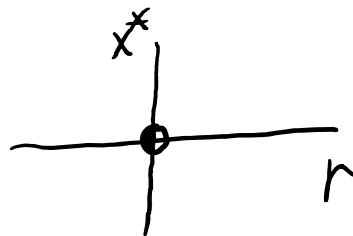
So, No saddle-node Bifurcation occurs

b) $\dot{x} = r^2 + x^2$



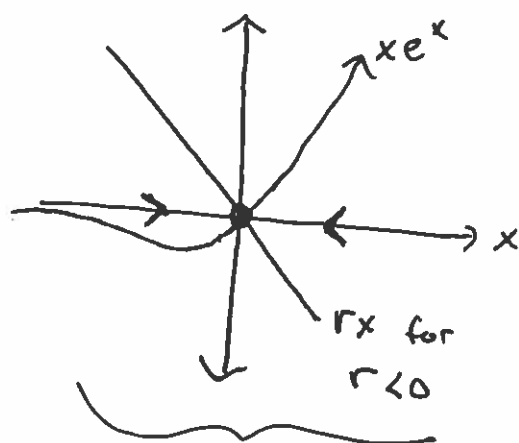
$$\dot{x} = 0 = r^2 + x^2$$

$$\Rightarrow x^* = 0$$

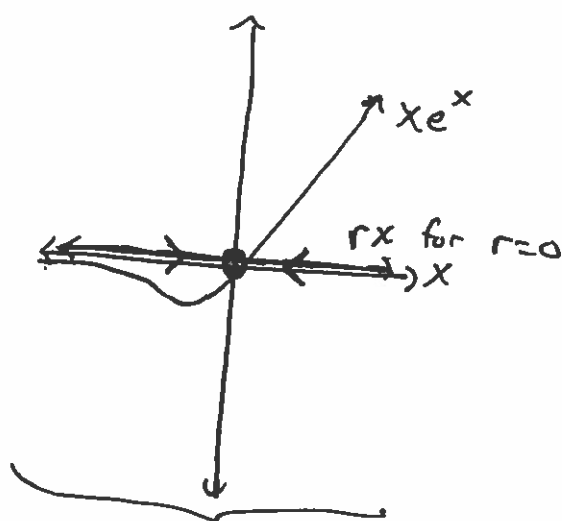


Similarly, Not a saddle-node bifurcation.

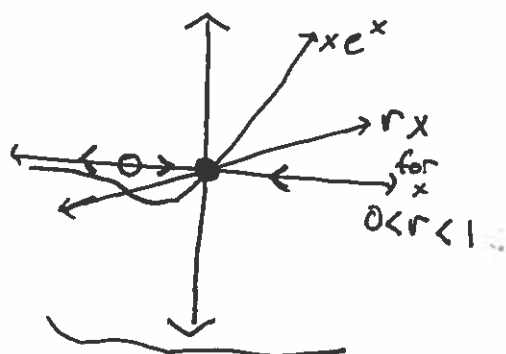
3.2.4 By inspection $x^* = 0$ is always a fixed point of $\dot{x} = x(r - e^x)$. We can see the different flows that occur for different values of r by plotting rx and xe^x for different values of r .



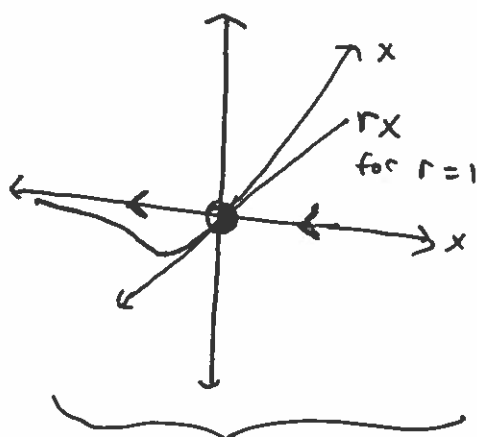
Only one fixed point at $x^* = 0$. It is stable



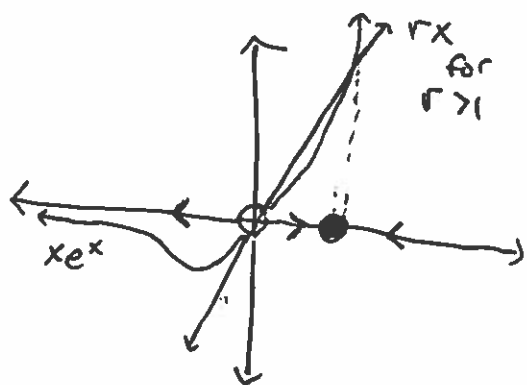
Fixed point appears at $x^* \rightarrow -\infty$



Two fixed points, One unstable ($x^* < 0$), one stable ($x^* = 0$)



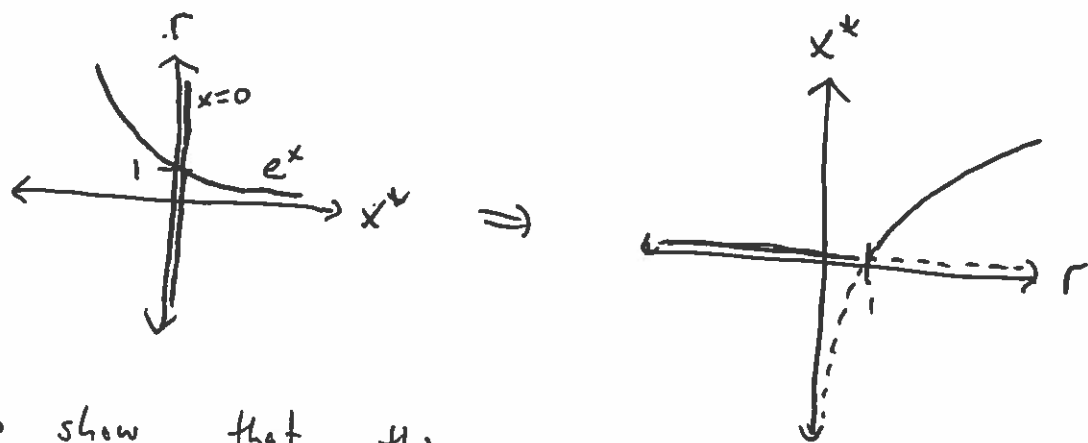
One bistable fixed point at $x^* = 0$.



We know the bifurcation must occur at $x=0$ (since x^* is always a fixed point. At bifurcation rx and $x e^x$ must be tangent so

$$\begin{aligned} \frac{d}{dx}[rx] \Big|_0 &= \frac{d}{dx}[x e^x] \Big|_0 \\ \Rightarrow r &= [e^x - x e^x] \Big|_{x=0} = 1 \\ \Rightarrow r_c &= 1 \end{aligned}$$

We can plot the bifurcation diagram by plotting r vs. x^* (i.e. $r(x^*) = e^{x^*}$) and then rotating the axes.



To show that this is indeed a transcritical bifurcation we can show that to leading order the Taylor expansion about the bifurcation point takes the form of the normal form for a transcritical bifurcation

$$x' = rx - x^2$$

$$\dot{x} \approx \underbrace{\dot{x}(x=0, r=1)}_0 + x \left. \frac{df}{dx} \right|_{x=0, r=1} + (r-1) \left. \frac{df}{dr} \right|_{x=0, r=1} + \frac{x^2}{2} \left. \frac{d^2f}{dx^2} \right|_{x=0, r=1} + x(r-1) \left. \frac{d^2f}{dx dr} \right|_{x=0, r=1}$$

$$= x \underbrace{(r - e^x - x e^x)}_0 \Big|_{x=0, r=1} + (r-1) \underbrace{(x)}_0 \Big|_{x=0, r=1} + \frac{x^2}{2} \underbrace{(-2e^x - x e^x)}_{-2} \Big|_{x=0, r=1} + \dots + \frac{(r-1)^2}{2} \left. \frac{d^2f}{dr^2} \right|_{x=0, r=1}$$

$$\dots + x(r-1)(1) + \frac{(r-1)^2}{2}(0)$$

$$= x \underbrace{(r-1)}_{r'} - x^2$$

Takes on the form $\dot{x} = rx - x^2$ near bifurcation point.

3.3.1

$$\dot{n} = GnN - kn$$

$$\dot{N} = -GnN - fN + P$$

a) if $\dot{N} \approx 0 \Rightarrow 0 \approx -GnN - fN + P \Rightarrow N \approx \frac{P}{Gn+f}$

so, $\dot{n} = \frac{GnP}{Gn+f} - kn$

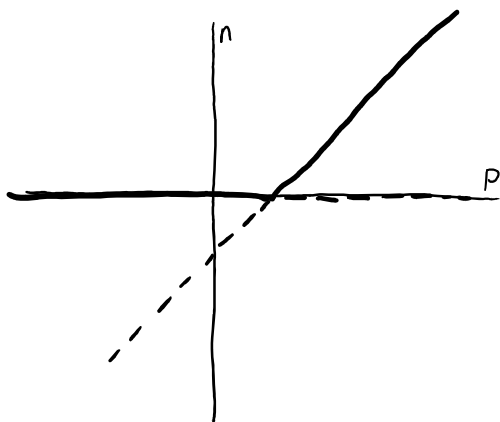
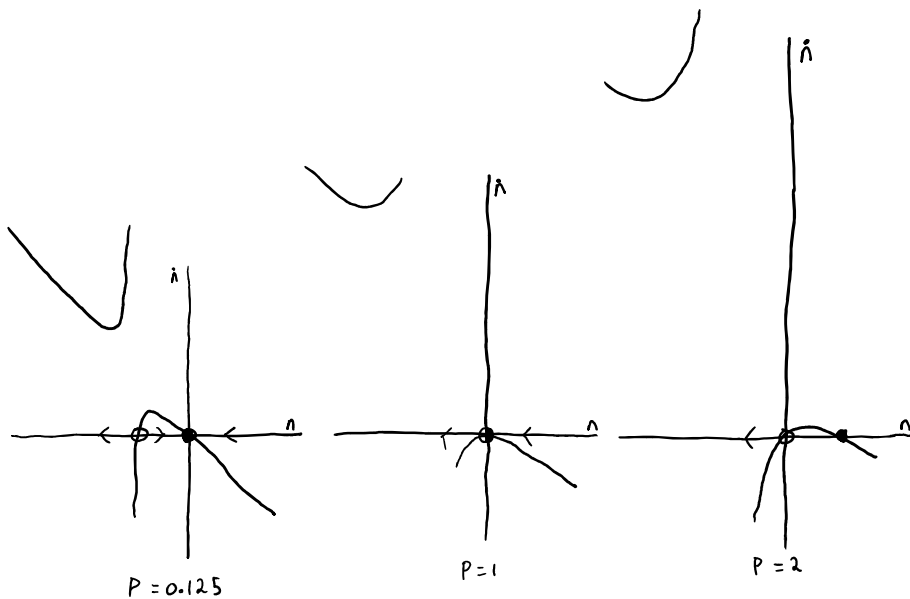
$$N = \frac{P}{Gn+f}$$

b) $\frac{d}{dn} \dot{n} = \frac{GP}{Gn+f} - \frac{G^2nP}{(Gn+f)^2} - k = \frac{fPG}{(Gn+f)^2} - k$

now look at $\left. \frac{d\dot{n}}{dn} \right|_{n=0} = \frac{PG}{f} - k$. for this to be unstable, $\frac{d\dot{n}}{dn} > 0$, so

$$\frac{PG}{f} - k > 0 \quad \text{or,} \quad P > \frac{kf}{G} = P_c$$

c) To investigate the bifurcation, let's set G, k , and f to 1 $\Rightarrow \dot{n} = \frac{Pn}{n+1}$
so, $n=0$ and $n=P-1$



so, This is a trans critical bifurcation

3.3.1 (cont.)

d) If $\dot{N} \approx 0$, N can be thought of as a constant. So, Let's look at $\dot{n}$ in this scenario

$$\dot{n} = (GN - K)n = \alpha n$$

This has an approx solution of

$$n(t) \approx n_0 e^{\alpha t}$$

Similarly,

$$\dot{N} = -(GN + f)N + P = -\beta N + P$$

$$N(t) \approx C_1 e^{-\beta t} + \frac{P}{\beta}$$

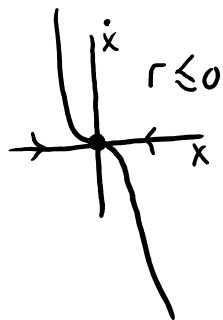
This means that as t evolves, these things race away from their values. So, for one of them to relax much faster than the other, we want $\beta \gg \alpha$ or,

$$GN + f \gg GN - K$$

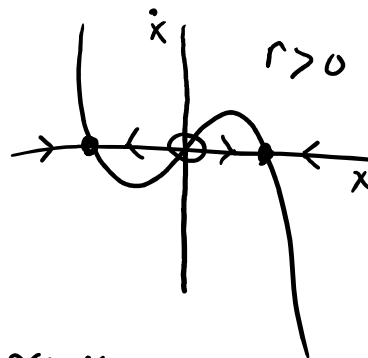
3.4.3

$$\dot{x} = rx - 4x^3$$

There are two qualitatively scenarios



and

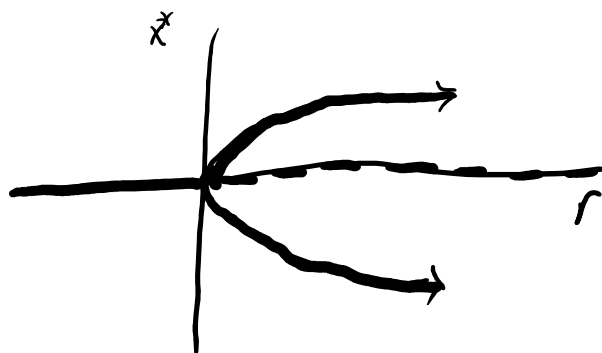


In order to analyze the bifurcation that occurs, we need to first find the fixed point solutions.

$$\dot{x} = 0 = rx - 4x^3 = x(r - 4x^2)$$

So, the solutions are $x^* = 0$ and $x^* = \pm\sqrt{\frac{r}{4}}$.

To gain intuition about the bifurcation, let's plot the solutions



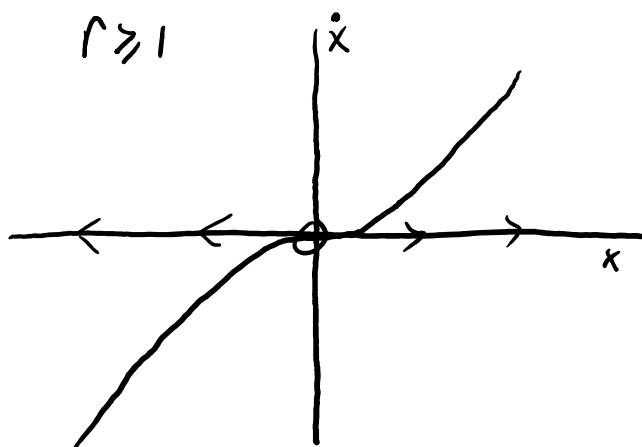
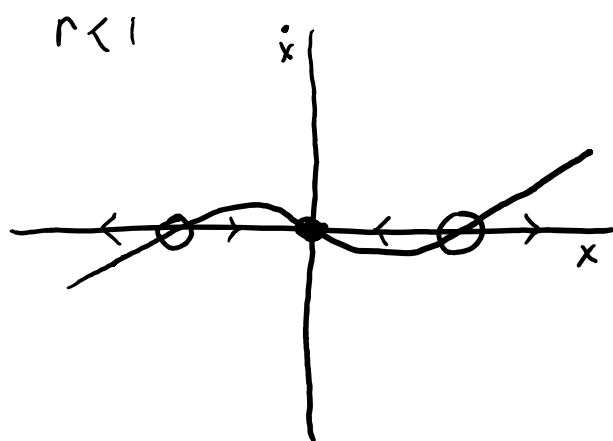
So, this is a subcritical pitchfork bifurcation at $r = 0$.

(it's subcritical because, as a function of r , the system would jump from one solution branch to another)

3.4.8

$$\dot{x} = rx - \frac{x}{1+x^2}$$

There are two qualitatively different scenarios:



To better understand what is going on, Let's find The Fixed Point solutions and Plot Them as a function of r .

$$\dot{x} = 0 \Rightarrow rx = \frac{x}{1+x^2}$$

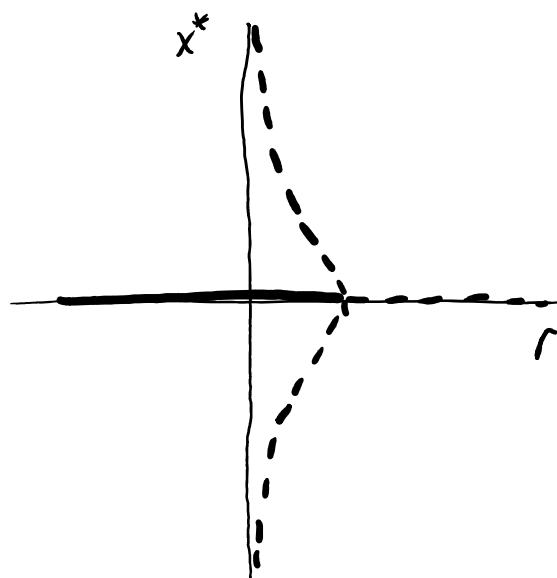
$$(1+x^2)rx = x$$

$x^* = 0$ is one solution

for other solutions

$$(1+x^2)r = 1$$

$$x^* = \pm \sqrt{\frac{1}{r} - 1}$$



So, This is a subcritical Pitch fork at $r=1$

3.5.7

a)

$$\dot{N} = rN \left(1 - \frac{N}{K}\right)$$

or,

$$\frac{dN}{dt} = rN - \frac{r}{K} N^2$$

$$\begin{aligned} [N] &\sim 1 \\ [t] &\sim t \end{aligned} \quad \text{so, } \left[\frac{dN}{dt}\right] \sim \frac{1}{t} \quad \text{and } \left[\frac{dN}{dt}\right] \sim [rN], \text{ so, } [r] \sim \frac{1}{t}$$

Similarly,

$$\left[\frac{r}{K} N^2\right] \sim \left[\frac{dN}{dt}\right], \text{ so } [K] \sim 1$$

b) Start by making the $\frac{N}{K}$ term dimensionless.

$$x = \frac{N}{K} \Rightarrow K \dot{x} = K r x (1-x), \text{ so } \frac{dx}{dt} = r x (1-x)$$

Now rescale time

$$\tau = r t \Rightarrow r \frac{dx}{d\tau} = r x (1-x) \quad \text{or, } \dot{x} = x(1-x)$$

c) To make the initial equal 1, we need $u = \frac{N}{N_0}$

$$\Rightarrow N_0 \dot{u} = r N_0 u \left(1 - \frac{N_0}{K} u\right)$$

again we can scale time as $\tau = r t$

$$\Rightarrow \frac{du}{d\tau} = u \left(1 - \frac{N_0}{K} u\right)$$

Now, This is dimensionless with a non-dimensional parameter $K = \frac{N_0}{K}$

$$\dot{u} = u(1 - Ku)$$

d) The non-dimensionalization with K leaves a 'knob' you can tune and study the effect of the carrying capacity. This non-dimensionalization is not sensitive to the initial condition though. So, to study the effect of the initial condition, you need to use the x non-dimensionalization